GGG-W-6lib
November 2, 1967
SUPERSEDING
Int. Fed. Spec. GGG-W-006lib (GSA-FSS)
November 1, 1965 and
Fed. Spec. GGG-W-6lb
June 23, 1961 (In part) (See 6.3)

FEDERAL SPECIFICATION

WRENCH, SOCKET (RIM NUT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal Agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers socket wrenches of the cross type and double end type (with bar handle) for installing and removing automotive rim nuts.

1.1.1 Federal specification coverage. Federal specifications do not include all varieties of the commodity as indicated by the title of the specification, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and sizes. Wrenches shall be of the following types:

Type I. Cross type (see fig. 1).
Type IV. Double end type (see fig. 2) for wrench and (see fig. 3) for handle. Handle may be procured separately.

Note: See 6.3 for supersession data covering types II and III wrenches.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitation for bids, or request for proposals, form a part of this specification:

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals: Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

Military Specifications:

MIL-H-15424 - Hand Tools, Packaging of.
MIL-P-16232 - Phosphate Coatings, Heavy Managanese or Zinc Base.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Atributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following publication forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

United States of America Standard Institute (USASI) Standard:
USASI B18.2-1960 Square and Hexagon Bolts and Nuts.

Application for copies should be addressed to the U. S. of American Standards Institute (USASI), 10 East 40th Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Material. Material shall be free from all defects and imperfections that might affect the serviceability of the finished product.

3.2 Design. The design shall be in accordance with the best engineering practice. Socket surfaces shall be well-defined.

3.3 Coating. Unless otherwise specified (see 6.2), the coating shall be in accordance with the manufacturers' commercial practice.

3.3.1 Phosphate coating. When specified (see 6.2), the wrenches and handles shall be coated in accordance with type II, class 2, of MIL-P-16232.

3.4 Dimensions and test loads. The distance across flats dimensions and test loads shall be in accordance with table I. The test load shall be applied as specified in 4.6.1.

TABLE I. Loads for testing

Nominal socket size (square or hex.)	Distance across flats	Socket test load
Inches	Inches	Inch-pounds
5/8	0.632 - 0.643	900
3/4	0.757 - 0.768	1300
13/16	0.820 - 0.831	1600
7/8	0.885 - 0.900	1900
15/16	0.948 - 0.965	2200
31/32	0.979 - 0.994	2300
1	1.010 - 1.025	2500
1-1/16	1.072 - 1.087	2800
1-1/8	1.135 - 1.150	3500
1-3/16	1.198 - 1.213	3800
1-1/4	1.260 - 1.275	4200
1-1/2	1.510 - 1.525	4700
1-3/4	1.760 - 1.775	7500

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

3.5 Type I wrench. Type I wrenches shall be the shape of a cross. The two cross bars shall be joined in such a manner that the four arms will be equal in length within 1/2 inch. The angle between any two adjacent arms shall be $90^{\circ} \pm 2^{\circ}$. Sockets shall be hardened to Rockwell C 39 minimum. Wrenches shall be of the size number specified (see 6.2 and table II), conform to table II, and be similar to figure 1.

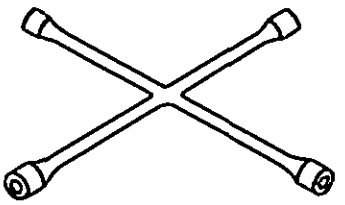


FIGURE 1. Type I, cross type wrench.

TABLE II. Type I, cross type wrench

Size	Nominal hexagon socket size	Nominal square socket size	Cross bar length (min.)
No.	Inches	Inch	Inches
1	7/8, 13/16, 3/4, 5/8	-	14
2	15/16, 7/8, 13/16, 3/4	-	15
3	15/16, 7/8, 13/16, 3/4	-	20
4	1-1/16, 1, 15/16, 7/8	-	20
9	1-1/2, 1-1/4, 1-1/8, 7/8	13/16, 5/8 1/	23
10	1-1/2, 1-1/4, 1-3/16	13/16	27

1/ The 13/16 square opening in the size 9 is located at the base of the 1-1/8 hexagon opening, and the 5/8 square opening is located at the base of the 7/8 hexagon opening.

3.6 Type IV wrench. Type IV wrenches shall be of one-piece construction with sockets at each end. Sockets shall be hardened to Rockwell C 28 minimum. Wrenches are intended to be used with the handles specified herein, shall conform to the dimensional requirements of figure 2 and table III for the size number specified, and be similar to figure 2.

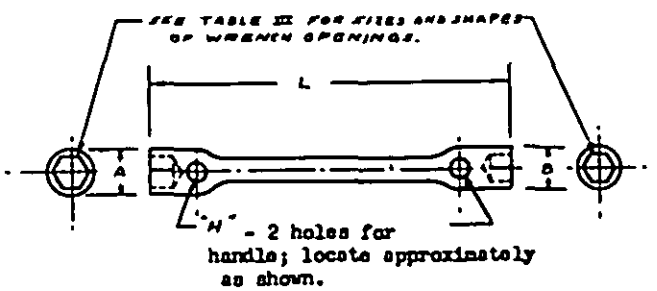


FIGURE 2. Type IV, double-end type wrench.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

TABLE III. Type IV double-end type wrench

Size	Nominal hexagon socket size	Nominal square socket size	Length	Hole Dia. + 1/64 -0	Large head	Small head	Suggested handle size No.
			L	H	A	B	
No.	Inches	Inch	(Min.) Inches	Inch	(Max.) Inches	(Max.) Inches	
1	15/16 - 7/8	-	14	25/32	1-1/2	1-3/8	1
2	1-1/16 - 7/8	-	13	25/32	1-5/8	1-3/8	1
3	1-1/8 - 1-1/16	-	16	25/32	1-5/8	1-5/8	1
4	1-3/16 - 1-1/16	-	16	25/32	1-3/4	1-5/8	1
5	1-1/4 - 1	5/8 1/	16	25/32	1-3/4	1-1/2	1
6	1-1/2 - 1-1/16	-	9	25/32	2-1/8	1-5/8	2
7	1-1/2 - 1-3/16	-	18	25/32	2-1/8	1-3/4	2
8	1-1/2	13/16	14	25/32	2-1/8	1-1/2	2
9	1-1/2	13/16	22	25/32	2-1/8	1-1/2	2
10	1-3/4	31/32	29	1-1/32	2-1/8	1-1/2	3

1/ The 5/8 square opening in the size 5 is located at the base of the 1-1/4 hexagon opening.

3.6.1 Handles. Handles shall be supplied as separate items or with type IV wrenches as required (see 6.2 and tables III and IV). The handles shall be hardened to Rockwell C 28 minimum, conform to the dimensions of figure 3 and table IV, and meet the test load of table IV.

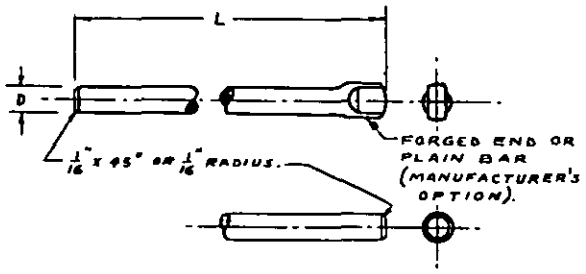


FIGURE 3. Handle for type IV wrench.

TABLE IV. Handle for type IV wrench

Size	Diameter + 1/64 -D	Length + 1/2 -L	Test load (Min.)
No.	Inch	Inch	Pounds
1	3/4	20	250
2	3/4	30	150
3	1	30	300

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:41Z
Check the source to verify that this is the current version before use.

3.7 Identification marking. Each wrench and handle shall be marked in a plain and permanent manner with the name or trademark of the manufacturer so that the source of manufacture may be readily determined.

3.8 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with good commercial practice. Burrs, sharp edges, flash, and other harmful or extraneous material shall be removed. Exposed edges shall be broken to 1/64 radius minimum.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table V.

4.3 Visual and dimensional examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, marking, and dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with 4.6.1 and 4.6.2.

4.5 Inspection of preparation for delivery. An inspection shall be made to determine that the packaging, packing, and marking requirements comply with section 5 of this specification. The inspection level shall be level II with an AQL of 4.0 defects per hundred units.

TABLE V. Sampling Data

Category	Sample unit	Inspection level	Acceptable quality level	AQL expressed in terms of	Reference
Visual examination	1	II	4.0	Defects per hundred units	3.1, 3.2, 3.3, 3.5, 3.6, 3.7, 3.8
Dimensional	1	II	4.0	Defects per hundred units	3.4, 3.5, 3.6
Testing	1	S-3	4.0	Defects per hundred units	3.4, 3.5, 3.6

4.6 Tests.

4.6.1 Test load. Each socket on all sample wrenches shall be tested. Prior to application of the load, the ends of the wrenches and handles shall be marked for later reference. The marking shall be such that distortion after the test shall be visible. Test mandrels shall be hardened to Rockwell C 55 minimum. Mandrels for sockets shall conform to (USASI) B.18.2-1960, for the "across the flats" dimensions and be of a length suitable to insure full socket engagement. When one end of a wrench contains two sockets, each socket shall be tested separately.

4.6.1.1 Handle test. Handles shall be tested by clamping one end in a vise for a distance of 6 inches. The load specified in table IV shall be applied to the opposite end of the bar at a distance of 12 inches from the vise jaws for the size 1 and 22 inches for sizes 2 and 3. Other methods of test satisfactory to the inspector may be used provided they apply a torque load equal to the product of the force and distance specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

4.6.1.2 Wrench test. Types I and IV wrenches shall be tested in a like manner. The type IV wrench shall have a bar of suitable strength inserted through the hole of the wrench farthest from the socket being tested. The wrench shall be placed on the test mandrel with the opposite end guided but free to rotate. Force shall be applied to the crossbar to produce the specified test load.

4.6.2 Hardness test. A hardness test shall be conducted in accordance with Federal Test Method Standard No. 151 to determine compliance with 3.5, 3.6 and 3.6.1.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. Unless otherwise specified (see 6.2), the wrenches shall be preserved, packaged and packed in accordance with MIL-H-15424 (as modified by 5.1.1) as specified for the applicable levels of preservation, packaging and packing.

5.1.1 As part of the requirements of section 5 of this specification, the following is considered as part of table I of MIL-H-15424:

Column 1	Column 2	Column 3	Column 7
Wrench; Socket:			
Cross			
6 through 10 pounds 15 oz.	P-1	I	12
11 through 14 pounds	P-1	I	6
Double end			
2-1/2 through 4 pounds 15 oz.	P-1	I	20
5 through 8 pounds 15 oz.	P-1	I	15
9 through 10 pounds	P-1	I	6
Handle	P-1	I	12

5.2 Marking

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The wrenches covered by this specification are intended for use on automotive rim nuts.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Type and size required (see 1.2.1 and tables II and III).
(c) If handles are required, state size number (see 3.6.1 and tables III and IV).
(d) If phosphate coating is required (see 3.3.1).
(e) Levels of packing and packaging required (see 5.1).
(f) State additional marking requirements, if required (see 5.2).

6.3 Type II bung wrenches of the non-sparking and nonmagnetic type formerly included in this specification are now covered by GGG-W-642.

Type II Bung wrenches (made of hardened steel) formerly included in this specification are now covered by GGG-W-647.

Type III lineman's wrenches formerly included in this specification are now covered by GGG-W-648.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T12:41Z
Check the source to verify that this is the current version before use.

MILITARY COORDINATING ACTIVITIES:

Army- GL
Navy-YD
Air Force-69

Review activities:

GL
YD
69

User activities:

WC, CE
MC
-

Preparing activity:

GSA-FSS

Orders for this publication are to be placed with General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price of this specification 5 cents each.

Source: https://assist.dla.mil -- Downloaded: 202640-26T12:41Z
Check the source to verify that this is the current version before use.